

PROBLEM SET 2

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 60)

- 1) (1) **(5 pts)** Show that the eigenvectors of $\boldsymbol{\sigma} \cdot \hat{\mathbf{n}}$ are given by Eq. (14.3.28).
 (2) **(5 pts)** Verify Eq. (14.3.29).

- 2) **(10 pts)** Express the following matrix M in terms of the Pauli matrices:

$$M = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$$

hint: use Eqs. 14.3.42, 14.3.43 where the indices run from 0 to 3

- 3) **(5 pts)** At $t = 0$, an electron is in the state with $s_z = \hbar/2$. A steady field $\mathbf{B} = B\mathbf{i}$, $B = 100$ G, is turned on. How many seconds will it take for the spin to flip ?

hint: a spin flip means a rotation of the spin of $\Delta\theta = \pi$; the average spin precesses around the $\mathbf{B}$ field at a frequency $\omega_i = -\gamma B_i$, $i = x, y, z$

- 4) (1) **(5 pts)** Estimate the relative size of the level splitting in the $n = 1$ state to the unperturbed energy of the $n = 1$ state, when a field $\mathbf{B} = 1000$ kG is applied.

(2) **(10 pts)** Recall that we have been neglecting the order B^2 term in H . Estimate its contribution in the $n = 1$ state relative to the linear $(-\boldsymbol{\mu} \cdot \mathbf{B})$ term we have kept, by assuming the electron moves on a classical orbit of radius a_0 . Above what $|\mathbf{B}|$ does it begin to be a poor approximation ?

- 5) (20 pts) A beam of spin-1 particles, moving along the y axis, is incident on two collinear SG apparatuses, the first with $\mathbf{B}$ along the z axis and the second with $\mathbf{B}$ along the z' axis, which lies in the x - z plane at an angle θ relative to the z -axis. Both apparatuses transmit only the uppermost beams. Show that the probability of observing the particles passing through the SG2 filter is given by $P = (1 + \cos \theta)^2/4$

hint: (a) spin-1 particles have an orthonormal basis of states $m = |1\rangle, |0\rangle, |-1\rangle$, (b) the uppermost beam refers to only the top half (above the y -axis only) part of the SG apparatuses, (c) the particles coming out of SG1 will be in a pure $S_z = +\hbar$ state $|1\rangle$, whereas the second SG2 apparatus will separate the beam according to $\mathbf{S} \cdot \hat{\mathbf{n}}'$, where $\mathbf{n}'$ is the unit vector along the z' axis; the uppermost part of the beam would then be $\mathbf{S} \cdot \hat{\mathbf{n}}' = +\hbar$ eigenstate $|1'\rangle$, and will need to compute $|1'\rangle$ in terms of the orthonormal basis as follows, $|1'\rangle = a|1\rangle + b|0\rangle + c|-1\rangle$ for which the coefficients will first need to be determined. Finally, one may obtain the probability of the particle beam passing through the second filter SG2 calculated as $P = |\langle 1'|1\rangle|^2 = |a|^2$

